

Sorting Without a Hat: Information Frictions and Early Academic-Track Choice in India*

Ayush Shahi

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Yale University

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Abstract

India's higher-secondary education system requires students to commit to a discipline at age 16, before they might know what they are good at. Does this early academic-track choice lead to talent misallocation? Using a panel dataset from the Consumer Pyramids Household Survey (2015–2025; ~40,000 individuals), I treat within-panel discipline switching as a revealed-preference proxy for initial mismatch and document three facts. First, women, rural households, and lower caste students switch substantially less than their counterparts, consistent with thinner information networks producing more rigid initial commitments and the high costs associated with switching. Second, within-household variation across siblings shows that having an elder sibling in the same field reduces switching by roughly six percentage points. Third, the cross-sectional wage premium for switchers, likely driven by unobservables, disappears under sibling fixed effects. Finally, I provide some suggestive evidence on policy shocks to the sorting margin by exploiting the national unification of India's medical entrance exam in 2017, which differentially affected the costs to STEM entry across states.

Keywords: discipline, sorting, misallocation, comparative advantage

*Reference to the sorting hat in Harry Potter.

1 Introduction

Choosing what to study is an important decision in people’s lives, given its consequences for their earnings and career outcomes. Even from the perspective of policymakers, enabling individuals to match with disciplines and careers in which they have a comparative advantage is beneficial for aggregate output— analogous to an argument made for ‘gains from trade’ (Hsieh et al., 2019).

What happens when these decisions are made with imperfect information about one’s ability? Young adults will sort into academic tracks inefficiently, with decisions influenced by other factors such as the perceived wage distribution of future occupations and parental advice (see Patnaik et al. (2020) and Altonji et al. (2012) for a review). A nascent literature has looked at the socio-economic inequality involved in this choice. Qiu (2025) uses administrative data from China to show that low-SES students are more likely to choose majors familiar to them, but which yield lower returns, compared to their high-SES peers due to information constraints. Aucejo et al. (2025) show similar frictions for first-generation college students in the US, with negative implications for their academic trajectory.

If people learn about their ‘true ability’ over time, there would be convergence in career outcomes for mismatched individuals through college major or job switching. However, in settings where educational investments are irreversible or the costs associated with switching are high, these inefficiencies will be exacerbated. For example, a student with comparative advantage as an economist might be stuck in an engineering occupation due to a decision made before their discipline-specific ability became clear.

This paper documents the decision of discipline choice, the conditions under which talent misallocation occurs, and its consequences for labor market outcomes of young adults. The setting of my study is India, which has a rigid educational structure that requires students to commit to an academic track at the age of sixteen. I treat within-panel discipline switching as a revealed-preference proxy for an initial mismatch and use three complementary identification strategies: a cross-sectional design that documents demographic correlates of switching, a

within-household design that uses elder-sibling discipline as a shifter of younger siblings’ priors, and some descriptive evidence on discipline choice induced by the 2017 unification of the National Eligibility cum Entrance Test (NEET) in India. Three findings preview the results. First, women, rural-household members, and lower-caste students switch substantially *less* than their counterparts, consistent with high costs to switching producing more rigid commitment to the initial choice for these groups. Second, having an elder sibling in the same aggregated field reduces own switching, both in the cross-section and within households once family fixed effects absorb shared unobservables. Third, the cross-sectional wage premium for switchers disappears under sibling fixed effects, suggesting that observed wage differences across switchers reflect positive selection rather than the causal premium to mid-stream revision.

The rest of the paper is organized as follows. Section 2 reviews the related literature on information frictions in major choice. Section 3 provides background on the institutional setting, including the the timing of stream choice. Section 4 describes the CMIE Consumer Pyramids data and the construction of the balanced panel and analysis sample. Section 5 outlines the empirical specifications. Section 6 presents the main results, and Section 7 discusses them. Section 8 concludes.

2 Background

India’s schooling system follows an 8+2+2 structure: eight years of elementary education, two years of secondary education, and two years of higher secondary education (corresponding to Grades 11 and 12 of US high school). At the end of secondary school (around age 16), students sit for “Board Exams” — administered by state, central, or private boards depending on school affiliation. Students who pass must then choose one of three broad academic tracks for higher secondary study: Science, Commerce, or Arts.

This discipline choice is consequential and costly to revise. The higher secondary curriculum for each track is distinct and non-overlapping: Science students study Physics, Chemistry,

and Mathematics (and optionally Biology for Medical students); Commerce students study Accountancy, Business Studies, and Economics; Arts students study History, Political Science, Literature, and related subjects. This specialization also determines access to university entrance examinations, which are typically track-specific and taken immediately after Class 12. For example, admission to the Indian Institutes of Technology requires clearing two stages of the Joint Entrance Examination, which tests exclusively Science-track material. Switching tracks mid-stream therefore requires catching up on a substantially different curriculum, foregoing the most direct route to track-specific entrance exams, and often renegotiating family expectations around career trajectories. These costs are unlikely to be uniform: students from lower-SES backgrounds plausibly face tighter financial constraints, less information about counterfactual returns, and stronger family pressure to honour the initial choice than their higher-SES peers, so the same realised mismatch may translate into very different switching probabilities across SES groups (Munshi and Rosenzweig, 2006).

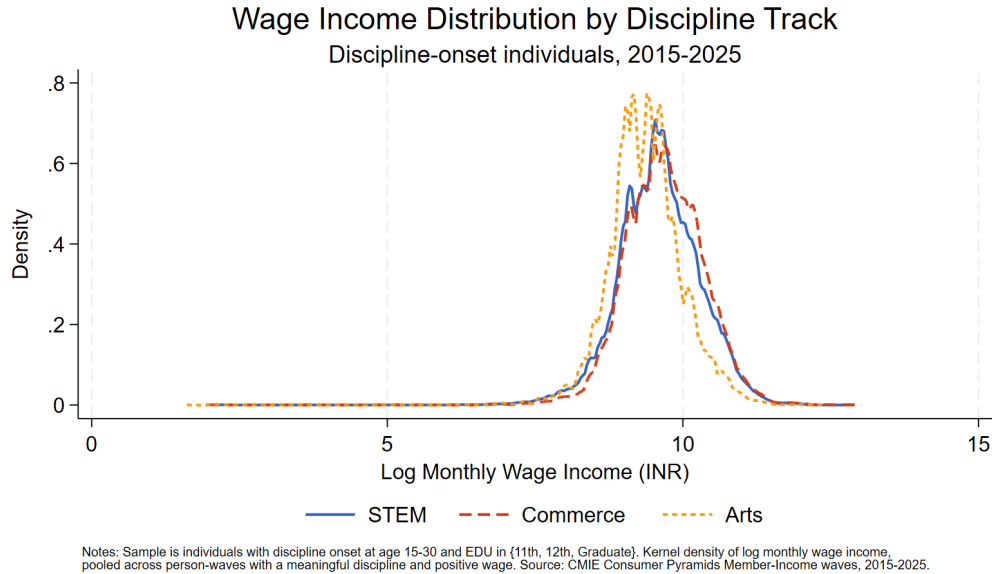


Figure 1: Kernel density of log monthly wage income by higher-secondary track.

Despite these institutional differences across tracks, the realized wage distributions look surprisingly similar. Figure 1 plots kernel densities of log monthly wage income by aggregated field, pooled across post-onset person-waves with positive wages. Although the popular

narrative around higher-secondary tracking is dominated by the Science premium, the three densities overlap substantially: the modes of all three lie close to one another, and the Arts density is shifted only modestly to the left, with a thicker lower tail. The visual similarity of the three distributions implies that average wage gaps across *aggregated* tracks understate the within-track variation that a mismatch story would emphasize. Whether the welfare cost of mis-sorting at age sixteen is large therefore depends not on these unconditional differences, but on how individuals would have fared under a counterfactual track.

3 Related Literature

A long literature documents that students choose college majors and high-school tracks under imperfect information about returns and about their own ability (Patnaik et al., 2020). Qiu (2025) uses Chinese administrative data to show that low-SES students disproportionately sort into majors that are familiar to them rather than those with the highest expected return, and that this gap is concentrated at the choice stage rather than reflecting differential ability. Aucejo et al. (2025) document an analogous friction for first-generation college students in the United States: lacking parental information about college, they enroll in lower-return majors and switch more often after matriculation. My paper extends this line of work to the Indian higher-secondary stream choice, which is more rigid than the US college-major choice.

The theoretical foundation for studying mismatch is the Roy (1951) model, in which agents with sector-specific productivity sort across occupations on the basis of comparative advantage. Static Roy models, however, assume that agents know their own productive type at the moment of choice. Two strands of work relax this assumption. Arcidiacono (2004) develops a dynamic model in which students hold imperfect priors about their ability, update upon observing noisy academic signals, and may switch majors when posterior beliefs revise sharply. Gauthier et al. (2023) estimate a generalised Roy model with dynamic correlated random coefficients, in which households learn about their relative productivity across sectors over time. I draw on this

framework conceptually – treating discipline switching as the empirical analogue of a posterior update – without estimating a structural model of my own. Misallocation in this view arises when a student with comparative advantage in (say) Arts sorts into Science under a noisy prior about their Science ability, perhaps anchored to the (perceived) large absolute wage premium of the Science track. The implication I take to the data is that misallocation should be *more* severe among lower-SES students, who have thinner family information networks for forming beliefs about within-track returns and a stronger incentive to chase the high-mean track.

Empirical evidence on the Indian context is sparse but growing. Jain et al. (2022) estimate that individuals who pursued Science in higher secondary earn up to 22% more than non-Science peers, but acknowledge that the relationship is not causal: ability, ambition, and parental investment plausibly drive selection into Science. Roychowdhury (2021) uses matching techniques and finds a more nuanced result – Science-track choice *reduces* employment probability but, conditional on employment, raises both earnings and the likelihood of professional occupation.

4 Data and Descriptives

For my analysis, I use the Consumer Pyramids Household Survey (CPHS), conducted by the Centre for Monitoring Indian Economy (CMIE) since 2014. The Income Pyramids module of the CPHS provides me with a decade of longitudinal data on education, employment, earnings, and other characteristics for roughly 170,000 households. Due to computational constraints, I assemble the panel at a quarterly, rather than monthly, frequency— which gives me 42 waves spanning March 2015 to September 2025.

Of interest to me is the discipline variable in CPHS, which specifies broad fields as well as specific professional courses. I aggregate into the three major disciplines as follows¹:

¹A residual “Others” bucket, with an insignificant N, contains disciplines that are either non-professional or not directly comparable to the main three streams. I drop individuals whose discipline ever falls in this bucket from the analysis sample.

- **STEM:** Science, Engineering/Polytechnic, Computer Applications, Medicine, Dentist, Pathology Lab Technician/DMLT, and Veterinary Science.
- **Commerce:** Commerce; Accounting/Tally/Taxation; Banking/Finance/Insurance; Management Studies; CA/CS/CFA/FRM (intermediate and completed).
- **Arts:** Arts, Fine Arts, Education, Law, Mass Media/Communication/Journalism, Architecture, Fashion / Interior / Apparel Design.

I restrict my panel to individuals who make their discipline choice during the time period of my panel. This excludes individuals who entered the panel already sorted into a stream. This is because I am interested in the question of early career and discipline sorting. As an additional check, I only include individual’s whose current education level is “11th Std. Pass”, “12th Std. Pass”, or “Graduate” and age is between 15 and 30, to ensure I am looking at individuals making the higher-secondary or college-entry stream choice.

These restrictions yield an analysis dataset of 40,249 individuals observed across 42 quarterly waves, for a total of roughly 1.7 million person-wave observations. Table 1 reports descriptive statistics by the aggregated field at the onset wave. Of my total sample, 31% choose Science (STEM), 11% Commerce, and 58% Arts. The three groups differ on observables in ways that suggest correlated SES selection. The share of rural households is highest among Arts choosers (44%), intermediate in STEM (34%), and lowest in Commerce (27%); the share of SC students follows the same ordering (25%, 17%, 15%); the share of female students is roughly stable at 39–41% across all three tracks. Conditional on positive wages, average monthly earnings are highest in Commerce and STEM, and notably lower in Arts. The wage and income figures are noisier than the demographic shares because positive-wage observations are relatively scarce in this young, mostly student sample.

Table 2 reports the first-to-last discipline transition matrix, where rows index each individual’s first meaningful discipline and columns index the last. Diagonal entries represent stability — individuals whose first and last discipline are the same — and off-diagonal entries represent

Table 1: Summary Statistics by Track

	Mean	SD	P25	P75	N
STEM					
Share of Rural Households	0.34	0.47	0.00	1.00	12,323
Share Female	0.40	0.49	0.00	1.00	12,323
Share SC	0.17	0.38	0.00	0.00	12,323
Avg Monthly Wage (INR)	16658.04	10906.15	9700.00	20250.00	2,969
HH Income at Choice (INR)	19025.55	20389.70	6500.00	25000.00	12,323
Commerce					
Share of Rural Households	0.27	0.44	0.00	1.00	4,573
Share Female	0.39	0.49	0.00	1.00	4,573
Share SC	0.15	0.36	0.00	0.00	4,573
Avg Monthly Wage (INR)	17518.17	12398.86	10200.00	20833.33	1,493
HH Income at Choice (INR)	17928.63	17352.36	6000.00	25000.00	4,573
Arts					
Share of Rural Households	0.44	0.50	0.00	1.00	23,353
Share Female	0.41	0.49	0.00	1.00	23,353
Share SC	0.25	0.43	0.00	0.00	23,353
Avg Monthly Wage (INR)	13634.07	8937.11	8528.57	16000.00	7,230
HH Income at Choice (INR)	14660.41	15835.94	385.00	20000.00	23,353
Total					
Share of Rural Households	0.39	0.49	0.00	1.00	40,249
Share Female	0.40	0.49	0.00	1.00	40,249
Share SC	0.21	0.41	0.00	0.00	40,249
Avg Monthly Wage (INR)	14897.94	10092.32	9000.00	17851.47	11,692
HH Income at Choice (INR)	16368.21	17638.49	4500.00	22000.00	40,249

Notes: Sample = individuals whose discipline turns on mid-panel, with onset age 15-30 and EDU in 11th, 12th, Graduate. Shares and demographics are measured at each individual's onset wave (one observation per individual) to avoid double-counting across the 42-wave panel. Avg monthly wage is the individual-level mean across meaningful positive-wage person-waves, then averaged across individuals within each panel.

Source: CMIE Consumer Pyramids Member-Income waves, 2015-2025.

at least one observed switch. Pooled across all starting tracks, just under 28% of individuals end the panel in a different aggregated field than they started — the within-panel switching margin that the empirical analysis below uses as a revealed-preference proxy for an initial mismatch.

Table 2: Discipline switching matrix

First discipline	Last discipline			
	STEM	Commerce	Arts	Total
STEM	7932 (64.4%)	1034 (8.4%)	3357 (27.2%)	12323
Commerce	839 (18.3%)	2264 (49.5%)	1470 (32.1%)	4573
Arts	2857 (12.2%)	1704 (7.3%)	18792 (80.5%)	23353
Total	11628	5002	23619	40249

Notes: Sample is individuals with discipline onset at age 15-30 and EDU in {11th, 12th, Graduate}. Each cell reports the count of individuals whose *first* meaningful discipline (row) and *last* meaningful discipline (column) take the indicated values, with the row percentage in parentheses (share of the row total).

Source: CMIE Consumer Pyramids Member-Income waves, 2015–2025.

5 Empirical Strategy

I take three complementary approaches to identifying misallocation in higher-secondary track choice and its consequences. First, I use discipline switching— the within-panel transition between aggregated fields — as a revealed-preference proxy for an initial mismatch. Second, I exploit within-household variation across siblings to shut down family-level unobservables that confound a pure cross-sectional comparison. Third, I exploit the 2017 nationwide unification of the medical-entrance exam (NEET) to provide some initial descriptive evidence on the Science-track sorting margin in the Indian state of Tamil Nadu.

All cross-sectional specifications collapse the panel to one observation per individual at the discipline-onset wave so that demographic shares are not double-counted across the 42-wave panel. Standard errors are robust unless otherwise noted; sibling-FE specifications cluster at the household.

5.1 Discipline Switching and Demographics

I begin with linear-probability and count regressions of the misallocation proxies on demographic correlates (Table 3):

$$S_i = \alpha + \beta' \mathbf{X}_i + \varepsilon_i, \quad (1)$$

where S_i is either $n_switches_i$ (the count of within-panel discipline switches) or any_switch_i (a 0/1 indicator for at least one switch); $\mathbf{X}_i$ contains a female dummy, a rural-household dummy, and a 4-level caste factor (OBC, SC, ST; Upper Caste omitted). Negative coefficients on rural and SC/ST indicators in Table 3 indicate that disadvantaged groups switch *less*, consistent with the Qiu (2025) hypothesis that lower-SES students have weaker information-gathering channels and therefore commit more rigidly to their initial choice.

5.2 Elder Siblings as Information Shifters

To isolate the effect of discipline choice on outcomes through the channel of comparative advantage allocation, I can exploit variation in the precision of students' prior beliefs at the time of choice. Previous work has documented that the presence and discipline choice of an *elder sibling* who has already passed through the higher secondary system acts as an exogenous shifter of the noise with which the younger sibling observes their comparative advantage (Altmejd et al., 2021). The logic is as follows: a younger sibling who observes an elder sibling navigate a particular track receives a noisy but informative signal about how members of their household perform in that track, which updates their prior about their own comparative advantage before making their own choice. The specification in Table 4 is:

$$S_i = \alpha + \beta_1 \text{Elder Sib}_i + \beta_2 (\text{Elder Sib}_i \times \text{Same Field}_i) + \gamma' \mathbf{X}_i + (\eta_h) + \varepsilon_i, \quad (2)$$

where $\text{Elder Sib}_i = 1$ if any other Son/Daughter in the household is older than i , and $\text{Same Field}_i = 1$ if at least one such elder sibling chose i 's aggregated field. Controls $\mathbf{X}_i$ are first meaningful

field, female, rural-HH, and the caste factor. Columns (1)-(2) of Table 4 estimate Eq. (2) on the cross-section of individuals; columns (3)–(4) add household fixed effects η_h , identifying the coefficients off within-household variation across siblings and clustering standard errors at the household level. The interaction β_2 tests whether information from a same-field elder sibling reduces the probability of switching after one’s own choice.

5.3 Wage Effects of Discipline Switching

To assess whether discipline switching carries a labour-market cost or premium, I regress the log of last-wave (2025 Q3) monthly wage on the *any_switch_i* indicator (Table 5):

$$\ln w_i = \alpha + \beta \text{any_switch}_i + \gamma' \mathbf{Z}_i + (\eta_h) + \varepsilon_i. \quad (3)$$

Column (1) is a cross-sectional OLS with controls $\mathbf{Z}_i$: first-meaningful-field dummies, female, rural-HH, and caste. Column (2) adds a sibling fixed effect (η_h at the household level), restricting the sample to households with at least two Sons or Daughters in the analysis sample so that within-HH variation in switching status identifies β ; standard errors cluster at the household. The within-sibling specification absorbs every time-invariant family characteristic (location, parental occupation, household income at school age, latent ability shared with siblings) that confounds the cross-sectional estimate.

5.4 Occupational Switching on Discipline Switching

Discipline switching may itself be observed but understate true mismatch if mismatched workers express their misallocation through post-education occupational churn rather than by re-declaring a discipline. Table 6 examines this directly:

$$O_i = \alpha + \beta S_i + \gamma' \mathbf{X}_i + \varepsilon_i, \quad (4)$$

where the outcome O_i is either $n_occ_switches_i$ or $any_occ_switch_i$ (constructed from consecutive-wave changes in the occupation variable in the data², and S_i is the corresponding discipline-switch proxy. Columns (1)–(2) use any_switch_i as the regressor; columns (3)–(4) use the count $n_switches_i$. A positive β indicates that discipline-switchers also exhibit greater occupational mobility – i.e., the two proxies move together rather than substituting for each other.

5.5 NEET Unification as a Natural Experiment

The fixed-effects and sibling-FE designs above identify within-family effects but cannot speak to whether *aggregate* sorting quality responds to information shocks. For that I exploit the 2017–18 unification of the National Eligibility cum Entrance Test (NEET) for medical-college admission, which raised the difficulty and standardised the signal of the medical pathway most sharply in Tamil Nadu, the only Indian state that previously admitted to medical colleges purely on the basis of Class 12 board marks. Karnataka serves as the control: it is geographically and demographically comparable but had operated a competitive state pre-medical exam (CET) since the 1980s, so the NEET transition was a marginal change there.

The analysis sample for this section restricts further to individuals who choose discipline in 11th grade in Tamil Nadu or Karnataka. Following the cohort timeline of the NEET legal history, I treat 2014–2015 entrants as clean pre-treatment, 2016 as partial anticipation (NEET restored in April 2016 with state-quota exemption), 2017 as the first enforcement year (legal uncertainty until August 2017), and 2018 onward as clean post-treatment. The post-NEET binary indicator is $choice_year \geq 2017$; $treat \times post$ is the standard difference-in-differences term.

Conditional on having sorted into the Science track, did the post-NEET cohort earn more and switch less? Table 7 reports:

$$Y_i = \alpha + \beta_1 post_i + \beta_2 TN_i + \beta_3 (post_i \times TN_i) + \delta' \mathbf{X}_i + \delta_t + \varepsilon_i, \quad (5)$$

²Excluding Student / Not Applicable / Data Not Available codes

restricted to Science choosers in Tamil Nadu and Karnataka. Outcomes Y_i are: log last-wave wage, discipline-switch count and indicator, and occupational-switch count and indicator. δ_t are cohort-year fixed effects; $\mathbf{X}_i$ contains female, age at choice, and individual income at choice. Standard errors cluster at the state.

6 Results

Table 3 reports the cross-sectional association between discipline switching and demographics. All coefficients are negative and statistically significant at the 1% level. Women switch 14.1 percentage points fewer disciplines on average than men in the count specification (column 1) and are 6.1 percentage points less likely to ever switch (column 2). Similarly, rural households exhibit a gap of a 9.4-percentage-point lower switching probability. The caste gradient is the steepest dimension: relative to the Upper Caste reference group, OBC, SC, and ST groups switch significantly less.

Table 4 estimates Eq. (2) under both cross-sectional OLS (columns 1–2) and a within-household specification with household fixed effects (columns 3–4). In the cross-section, an individual whose elder sibling chose a *different* aggregated field is roughly 3 percentage points more likely to switch, while one whose elder sibling chose the *same* field is approximately 3 percentage points *less* likely to switch. Within siblings of the same household, the same-discipline elder-sibling channel reduces switching by 1.5 percentage points.

Table 5 regresses log monthly wage in the panel’s last wave on the ever-switched indicator. Column 1 is a cross-sectional OLS with the standard control set; column 2 is a sibling-fixed-effects specification restricted to households with at least two Sons or Daughters in the analysis sample. In the cross-section, switchers earn 6.8% more than non-switchers (significant at 5%). Once household fixed effects absorb shared family characteristics, the point estimate falls to -0.006 and is no longer statistically distinguishable from zero.

Figure 2 plots mean log monthly wage by single-year age, separately for discipline switchers

Table 3: Misallocation proxies on demographics

	(1) # of switches	(2) Any switch
Female	-0.141*** (0.010)	-0.061*** (0.005)
Rural HH	-0.191*** (0.010)	-0.094*** (0.005)
<i>Caste category:</i>		
OBC (vs UC)	-0.168*** (0.013)	-0.064*** (0.006)
SC (vs UC)	-0.274*** (0.014)	-0.129*** (0.007)
ST (vs UC)	-0.288*** (0.022)	-0.136*** (0.012)
Constant	0.867*** (0.012)	0.492*** (0.006)
Observations	35,265	35,265
R-squared	0.032	0.026

Notes: Outcome in (1) is the within-panel count of discipline switches. Outcome in (2) is a 0/1 indicator for any switch. Sample = individuals with discipline onset at age 15-30 and EDU in 11th, 12th, Graduate, collapsed to one observation per individual at the discipline-onset wave. Standard errors in parentheses. Stars: * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Source: CMIE Consumer Pyramids Member-Income waves, 2015-2025.

Table 4: Discipline switches and sibling effects

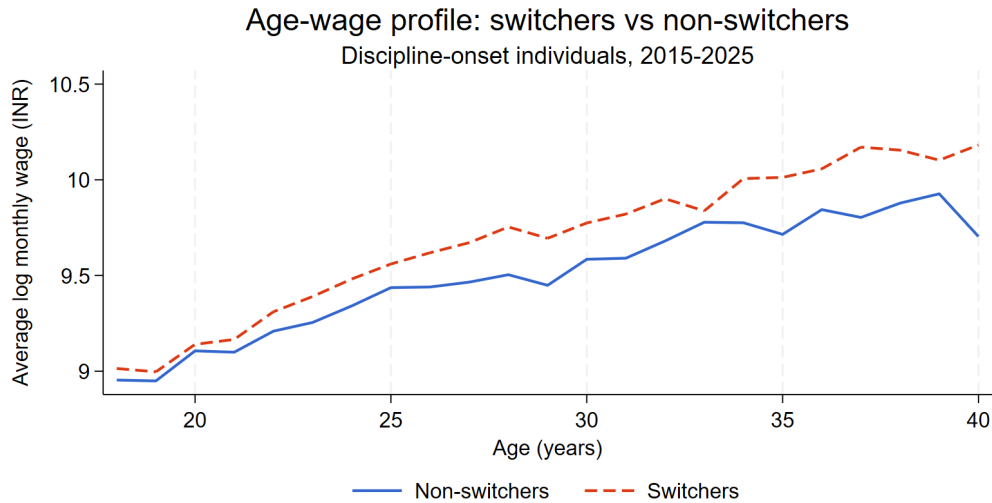
	Individual: Cross-section		Individual: Panel	
	(1)	(2)	(3)	(4)
	# of switches	Any switch	# of switches	Any switch
Has elder sibling	0.059*** (0.009)	0.030*** (0.004)	-0.087*** (0.014)	-0.020*** (0.006)
Elder sibling x same discipline	-0.147*** (0.012)	-0.061*** (0.006)	-0.037** (0.015)	-0.015** (0.007)
Controls	Yes	Yes	Yes	Yes
Household FE	No	No	Yes	Yes
Observations	40,545	40,545	1,702,890	1,702,890
R-squared	0.296	0.423	0.813	0.851

Notes: Columns (1) and (2) use one observation per individual taken at that individual's discipline-onset wave. Columns (3) and (4) use the full panel (42 waves) with household fixed effects absorbed and standard errors clustered at the household. Controls are first meaningful field, female, rural-HH, and SC/ST dummies, all measured at the onset wave and held constant across waves.

Source: CMIE Consumer Pyramids Member-Income waves, 2015-2025.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

and non-switchers, restricted to ages 18–40 and post-onset person-waves with positive wages. Wages rise with age as expected, with the lines diverging slightly in the early thirties—switchers sitting modestly above non-switchers, consistent with the 0.068 cross-sectional coefficient in Table 5.



Notes: Sample = positive-wage person-waves at ages 18–40 from individuals with discipline onset at age 15–30 and EDU in {11th, 12th, Graduate}. Switcher = 1 if the individual ever changed aggregated field across waves ($n_switches \geq 1$). Each point is the mean log wage in the cell; cells with fewer than 30 person-wave observations are dropped.
Source: CMIE Consumer Pyramids Member-Income waves, 2015-2025.

Table 5: Last-wave wage and discipline switching

	(1) Cross-section	(2) Sibling FE
Ever switched discipline (0/1)	0.068** (0.027)	-0.006 (0.080)
<i>Field:</i>		
Commerce (vs STEM)	0.007 (0.026)	
Arts (vs STEM)	-0.135*** (0.028)	
Female	-0.262*** (0.045)	-0.125 (0.202)
Rural HH	-0.274*** (0.017)	
<i>Caste Category:</i>		
OBC (vs Upper Caste)	-0.140*** (0.024)	
SC (vs Upper Caste)	-0.167*** (0.024)	
ST (vs Upper Caste)	-0.352*** (0.036)	
Age at choice		0.016 (0.012)
Constant	10.083*** (0.030)	9.451*** (0.255)
Observations	4,425	2,673
HH (sibling) clusters		2,218
R-squared	0.133	0.962

Notes: Outcome is log monthly wage in the panel's last wave (2025-09); individuals without a positive wage in that wave drop out of the regression sample. Column (1): cross-sectional OLS with controls from first meaningful field. Column (2): sibling fixed-effects – restricted to households with atleast two children. SE clustered at the household. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: CMIE Consumer Pyramids Member-Income waves, 2015-2025.

Table 6 examines whether the two switching margins move together. Columns 1–2 use the binary discipline-switch indicator as the regressor; columns 3–4 use the count. Discipline switchers are 14.8 percentage points more likely to ever switch occupations than non-switchers (column 1), and exhibit 1.045 additional occupational switches over the panel on average (column 2). When the count of discipline switches is the regressor, each additional discipline switch is associated with a 7.6 percentage-point increase in the probability of ever switching occupations (column 3), and 0.63 additional occupational switches (column 4). All four coefficients are significant at the 1% level.

Table 6: Occupational switching on discipline switching

	(1) any_occ_switch	(2) n_occ_switches	(3) any_occ_switch	(4) n_occ_switches
Ever switched discipline (0/1)	0.148*** (0.007)	1.045*** (0.059)		
Number of discipline switches			0.076*** (0.003)	0.630*** (0.034)
Controls	Yes	Yes	Yes	Yes
Observations	35,265	35,265	35,265	35,265
R-squared	0.055	0.095	0.057	0.099

Notes: Outcomes are individual-level occupational-switching measures. any_occ_switch = 1 if the individual ever changed occupation between two consecutive panel waves (excluding Student / Not Applicable / Data Not Available codes); n_occ_switches counts such transitions over the 42-wave panel. Cols (1)-(2) regress on any_switch (binary discipline switching); cols (3)-(4) regress on n_switches (count of discipline switches). Sample = individuals with discipline onset at age 15-30 and EDU in 11th, 12th, Graduate, collapsed to one observation per individual at the discipline-onset wave. SE clustered at the household. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: CMIE Consumer Pyramids Member-Income waves, 2015-2025.

Table 7 restricts the sample to Science choosers in Tamil Nadu and Karnataka and reports a difference-in-differences specification on five outcomes. The DiD coefficient on $Post \times High-shock$ is the parameter of interest in each column. On log wage (column 1), the point estimate is 0.167 but the sample of 68 observations is too small to support meaningful inference. On the count of discipline switches (column 2), the DiD coefficient is 0.285 and is significant at the 10% level; the corresponding binary indicator (column 3) is -0.092 and not significant. On occupational switching, both the count and indicator coefficients are small and not significantly

different from zero.

Table 7: Outcomes for Science choosers (post-NEET sorting)

	(1) ln(wage)	(2) n_disc_switches	(3) any_disc_switch	(4) n_occ_switches	(5) any_occ_switch
Post NEET ($i=2017$)	-0.324 (0.085)	-1.122* (0.174)	-0.420 (0.167)	-0.745 (0.288)	-0.210 (0.060)
High-shock state (TN)	-0.414** (0.029)	-1.107*** (0.014)	-0.528*** (0.001)	0.019 (0.065)	-0.047*** (0.000)
Post x High-shock	0.167 (0.059)	0.285* (0.028)	-0.092 (0.016)	0.012 (0.058)	0.006 (0.002)
Female	0.127 (0.076)	-0.129* (0.013)	-0.063* (0.007)	-0.677 (0.139)	-0.093 (0.032)
Age at choice	0.001 (0.008)	-0.024 (0.005)	-0.013** (0.001)	0.224* (0.032)	0.028*** (0.000)
Income at choice	-0.000** (0.000)	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	0.000*** (0.000)
Constant	10.266*** (0.098)	2.286*** (0.028)	1.204** (0.033)	-2.336* (0.259)	-0.015 (0.049)
Observations	68	1,885	1,885	1,885	1,885
R-squared	0.417	0.273	0.315	0.205	0.102

Notes: Sample restricted to individuals who chose Science (field_onset = STEM) with onset EDU = 11th Std. Pass in Tamil Nadu (treated) or Karnataka (control). Outcomes: (1) log wage in the panel's last wave (drops zero-wage observations); (2)-(3) discipline-switching count and indicator (within-panel); (4)-(5) occupational-switching count and indicator (consecutive non-Student non-NA occupations). Cross-section, one observation per individual at onset. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Source: CMIE Consumer Pyramids Member-Income waves, 2015-2025.

7 Discussion

The empirical patterns documented in Tables 3–7 admit several overlapping interpretations rather than a single clean story. The negative demographic gradient in Table 3 — women, rural households, and lower-caste students switching consistently less than their counterparts — is consistent with the Qiu (2025) information channel: households with thinner informational networks may anchor more rigidly on whatever stream they initially commit to and revise less

frequently as new signals arrive. But the same pattern is also consistent with at least two non-information mechanisms. First, lower-resource households may face higher *pecuniary* switching costs (forgone tuition, family support, loss of the cohort) and therefore fail to act on revised priors even when they exist. Second, lower-resource households may face stronger *family-pressure* costs: in cultures where parental approval of the chosen track is binding, mid-stream revision may not be a feasible option for the student. These three channels — information, financial constraints, and family pressure — are observationally equivalent in Table 3 and cannot be separated without a sharper exogenous shock to one of them.

Tables 4 and 5 similarly suggests that household-level selection drives much of the cross-sectional pattern: families that produce elder siblings observed in the panel differ systematically from those that do not, and these differences correlate with switching. Net of family unobservables, having a same-field elder sibling reduces own switching, which is the information-spillover prediction. Table 5 shows that the 6.8% cross-sectional wage premium for switchers vanishes under sibling fixed effects, indicating that the observable wage gap reflects positive selection into switching — by ability, ambition, or resources — rather than the causal return to mid-stream revision.

Table 6 documents that discipline and occupational switching move strongly together. The pattern is compatible with two stories. One is a “type” story: exploratory individuals churn across many margins of choice, and the two switching measures pick up the same trait. The other is a “mismatch propagates” story: an initial mistake at the discipline stage forces an extended search for a matching occupation. I cannot separate these two explanations; doing so would require an exogenous shock to one switching margin.

Table 7’s NEET specifications speak to whether a “cost shock” to the Science track— in terms of preparation required— changed sorting outcomes for those who chose Science in Tamil Nadu. Since my sample for this analysis is underpowered, the point estimates are mostly insignificant. In future work, I’ll try to think more carefully about the appropriate control group and the identification strategy.

8 Conclusion

This paper studies misallocation in India’s higher-secondary tracking system, treating within-panel discipline switching as a revealed-preference proxy for an initial mismatch. I find that lower-SES students switch substantially less than their counterparts and that the cross-sectional wage premium for switching is primarily driven by household-level positive selection.

The policy implications of my work relates to why disadvantaged groups switch less. If imperfect information at age sixteen is the binding channel, then targeted career counselling, exposure programmes, and clearer dissemination of within-track returns — especially in rural and lower-caste schools — could reduce mismatch at low cost. If switching costs (financial, temporal, or family-pressure-driven) instead bind, structural reforms that lower the cost of mid-stream revision — bridge curricula, more flexible track-specific entrance exams, and lower-cost retracking pathways — would matter more. Future work should look at disentangling these three mechanisms — imperfect information, constraints on revision, and direct preference for the initial track — by using an instrument or policy shock that shifts only one channel. This paper provides background work for the above, documenting who switches, when, and at what cost.

References

- Altmejd, A., A. Barrios-Fernández, M. Drlje, J. Goodman, M. Hurwitz, D. Kovac, C. Mulhern, C. Neilson, and J. Smith (2021, 08). O brother, where start thou? sibling spillovers on college and major choice in four countries*. The Quarterly Journal of Economics 136(3), 1831–1886.
- Altonji, J. G., E. Blom, and C. Meghir (2012, April). Heterogeneity in human capital investments: High school curriculum, college major, and careers. Working Paper 17985, National Bureau of Economic Research.
- Arcidiacono, P. (2004). Ability sorting and the returns to college major. Journal of Econometrics 121(1), 343–375. Higher education (Annals issue).
- Aucejo, E. M., J. French, P. Ugalde Araya, and B. Zafar (2025, August). Understanding gaps in college outcomes by first-generation status. Working Paper 34129, National Bureau of Economic Research.
- Gauthier, J.-F., T. Molina, and A. Nyshadham (2023). Learning and selection in the sorting of households across sectors. Working paper.
- Hsieh, C.-T., E. Hurst, C. I. Jones, and P. J. Klenow (2019). The allocation of talent and u.s. economic growth. Econometrica 87(5), 1439–1474.
- Jain, T., A. Mukhopadhyay, N. Prakash, and R. Rakesh (2022). Science education and labor market outcomes in a developing economy. Economic Inquiry 60(2), 741–763.
- Munshi, K. and M. Rosenzweig (2006, September). Traditional institutions meet the modern world: Caste, gender, and schooling choice in a globalizing economy. American Economic Review 96(4), 1225–1252.
- Patnaik, A., M. J. Wiswall, and B. Zafar (2020, August). College majors. Working Paper 27645, National Bureau of Economic Research.

Qiu, X. (2025). Choosing what's familiar: Information inequality in college major choices. Working paper.

Roy, A. D. (1951). Some thoughts on the distribution of earnings. Oxford Economic Papers 3(2), 135–146.

Roychowdhury, P. (2021). (em)powered by science? estimating the relative labor market returns to majoring in science in high school in india. Economics of Education Review 82, 102118.